مدارس الإمارات الوطنية
Emirates National Schools

Grade: 6 | Term 1, Week 13 (23 Nov to 27 Nov) | TCD: Future Skills 6: Applied Technology Foundations | Session length: 2 hrs

T1 WEEK: 13	**Lesson 1: Selection and Loops Mission**
Standard(s):	Technology: selection and repetition in algorithms, debugging; Math: 6.EE.B (reasoning with variables); ELA: SL.6.1 (collaborative discussion)
Literacy Strategies:	Condition frame: If Photon senses ..., then it ..., else it ...; troubleshooting log with sentence starters.
Assessment:	Mission success; troubleshooting log with two errors; peer debugging notes; live quiz on the TCD website (counts towards the Photon AI Mission certificate).
Lesson Objectives:	We are learning to use if-then conditions and loops so a robot can make decisions and repeat actions.
Success Criteria:	I can explain selection and repetition. I can program Photon to react when its sensor detects an obstacle. I can use a loop so my program keeps checking the sensor. I can log at least two errors and fix them, including one found by a peer.
Targeted Vocabulary	selection, condition, if-then-else, repetition, loop, sensor, obstacle, bug, optimise
Resources	Photon robots, charged, Tablets with the Photon coding app, Mission mats, Obstacle blocks, Troubleshooting log sheets, Flowchart symbol cards Accompanying resource: Grade6_Week13_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade6_Week13_Worksheet and Grade6_Week13_Lesson_Plan.
National Identity Integration	Link to driverless vehicles being tested in Dubai and Abu Dhabi, which use sensors and if-then decisions to avoid obstacles safely.
AI/Digital Learning	Photon robots and tablets with the Photon coding app; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Teach selection with flowchart diamonds (slide 6). Teach loops and why the condition goes inside the loop (slide 7). Model troubleshooting with a deliberate bug (slide 8).
Guided Practice: We Do	Class draws one obstacle flowchart together. Pairs code a single obstacle check and test it (slide 10).
Independent Practice: You Do	Pairs complete the obstacle mission and keep the troubleshooting log (slide 12). Peer debugging swap for 10 minutes (slide 9). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: Beginning / On / Above	**Beginning: Use the flowchart template and code one obstacle check inside a loop.** **On: Complete the full obstacle mission with a working if-then-else inside a loop.** **Above: Add a second condition (for example a sound or light reaction) and explain how your optimisation made the program better.**
Closure/Reflection	Pairs share the best bug they found and how they fixed it. Exit ticket (slide 15). Robots back to charging.
Inclusion / EAL Support	☑ Visual Supports ☐ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☐ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☑ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☑ Scaffold / Sentence Starters

مدارس الإمارات الوطنية
Emirates National Schools

G&T / Extension	☑ Extension / Challenge Task ☐ Real-World Application ☑ Higher-Order Questioning ☐ Curriculum Compacting ☐ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☑ Leadership / Peer Coaching ☐ Open-Ended Task ☐ Other: _____